1st AIAA CFD Transition Modeling and Prediction Workshop: OVERFLOW Results for the SST and Langtry-Menter Models

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This paper reports the results of CFD simulations of the test cases for the AIAA 1st CFD Transition Modeling and Prediction Workshop held at the AIAA Scitech Forum on 21-22 January, 2021 using the NASA solver OVERFLOW. The Langtry-Menter transition model was used in combination with the Shear Stress Transport (SST) turbulence model. The test cases include the ERCOFTAC zero-pressure-gradient flat plate cases T3A and T3B with by-pass transition, the natural laminar flow NLF(1)-0416 airfoil, the DLR inclined 6:1 prolate spheroid, and the NASA Common Research Model with wing design optimized for natural laminar flow transition (CRM-NLF). OVERFLOW load and transition predictions obtained using best practices are compared with experiments and prior simulations. Additional results are provided for modified turbulence model boundary specifications to improve alignment to reference data and assess model sensitivity.

I. Nomenclature

c	=	Chord length
c_d	=	Drag coefficient
c_f	=	Skin friction coefficient
c_l	=	Lift coefficient
c_p	=	Pressure coefficient
$C_{\epsilon 2}$	=	Coefficient in the destruction term of the turbulence dissipation equation
k	=	Turbulence kinetic energy
L	=	Reference length
M	=	Mach number
MUTINF	=	Eddy to kinematic viscosity ratio $\nu_{t\infty}/\nu_\infty$
N	=	Number of grid nodes
Re_1	=	Reynolds number per grid unit
Re_x	=	Streamwise running Reynolds number
$Re_{\theta t}$	=	Local transition onset momentum-thickness Reynolds number
S	=	Wing span
Tu	=	Turbulence intensity $\sqrt{2k/3}/u$
u_∞	=	Freestream velocity
$x,\ y,\ z$	=	Cartesian coordinates
β^*	=	Coefficient in turbulence kinetic energy dissipation term
γ	=	Intermittency
η	=	Fraction of span y/S
ϕ	=	Angular coordinate around the prolate spheroid longitudinal axis
ρ	=	Density
μ	=	Molecular viscosity
ν	=	Kinematic viscosity
ν_t	=	Eddy viscosity
ω	=	Specific turbulence dissipation rate

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II. Introduction

The 1st AIAA CFD Transition Modeling and Prediction Workshop was organized in an effort to assess the current state of the art in laminar-turbulent transition prediction in an industrial CFD environment [1]. The primary objectives of the event were to test best practices for transitional flow simulations, and to verify transition and turbulence model implementations. Data submitted by 18 participants from government, industry and academic organizations were compiled and compared. All participants used different combinations of models and model options, so that strict verification was not possible based on these results. Among the models used, intermittency-based models were the most frequently used. The core models included $\gamma - \mathrm{Re}_{\theta t}$ models such as the Langtry-Menter model [2] coupled with the SST [3] or Spalart-Almaras (SA) [4] turbulence model, the Menter γ model with SST [5], the Amplification Factor Transport model with SA [6], linear stability models coupled to SA or SST, the Medida-Baeder model with SA [7], the Bas-Çakmakçioğlu model with SA [8] and the Lag Elliptic Model [9]. Models accounting for crossflow induced transition were used by some of the participants. The results showed a wide spread and large deviations from reference data, indicating a lack of consistency in current transition prediction capabilities.

In the present work, Reynolds-Averaged Navier-Stokes (RANS) simulation results obtained with the OVERFLOW solver [10] are reported for the Langtry-Menter model with optional extension for crossflow transition [11], coupled with the SST model. The paper addresses an issue that was identified during the simulation exercise with the turbulence index formulation used to detect the transition front, as well as the current lack of transition specific best practices to set up the turbulence model farfield boundary conditions. The paper is organized as follows: the workshop test cases, including flow conditions, geometry and computational grids are described in Section III. Information on the solver, selected numerical methods and models is provided in Section IV. In Section V, the results obtained for each test case are presented and discussed. Section VI summarizes case-specific findings and more general observations on model performance.

III. Workshop cases

The workshop cases are summarized in Table 1. They include a grid refinement study for the European Research Community on Flow Turbulence and Combustion (ERCOFTAC) T3A and T3B flat plate cases tested by Roach and Brierley [12]; a grid refinement study at 0° and 5° angle of attack, and an angle of attack sweep for NASA NLF(1)-0416 general aviation airfoil [13]; mesh refinement and angle of attack variations for the inclined 6:1 prolate spheroid measured at the DFVLR 3x3 low speed wind tunnel facility in Göttingen, Germany [14], which represents a simplified fuselage configuration; and finally mesh refinement and angle of attack variations for the NLF design of the NASA Common Research Model (CRM-NLF) [15, 16].

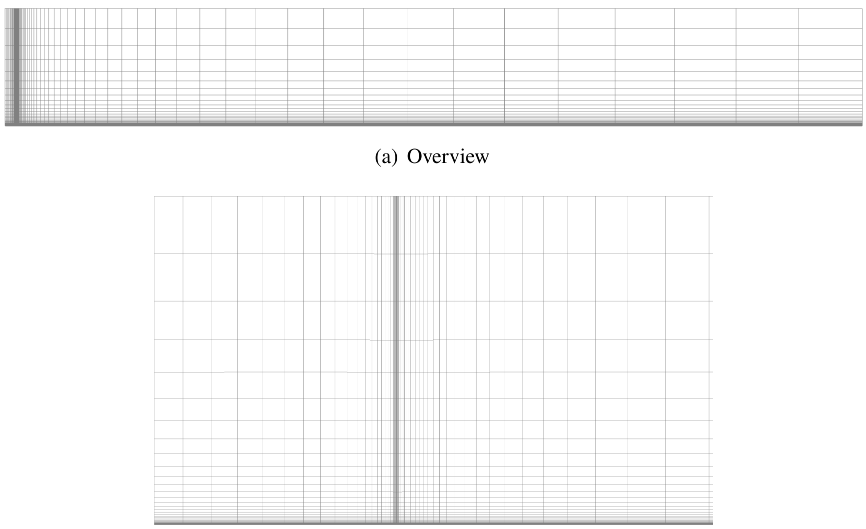
Table 1 Test Cases

Case	Description	Angle of Attack α	M	Re_1	Tu[%]
1A-B	2D zero-pressure-gradient flat plate, T3A and T3B	n/a	0.2	2×10^5	5.855 (A), 7.216 (B)
2A-C	2D NLF(1)-0416	$-4° \rightarrow 8°$	0.2	4×10^6	0.15
3A-B	3D Inclined, 6:1 prolate spheroid	$5°, 10°, 15°$	0.13	6.5×10^6	0.15
4A-B	3D CRM-NLF	$1.45°, 1.98°, 2.46°, 2.94°$	0.856	1.04×10^6	0.24

The structured grids made available to the workshop participants were used [1]. Five grid refinements were made available for the flat plate and prolate spheroid, 6 grids were provided for the NLF (1)-0416 airfoil, and 4 overset grid systems were prepared for the CRM-NLF model. Views of the coarse grid from each series are shown in Figs. 1-4. The number of grid points in each grid is given in Table 2. The flat plate is 20 grid units long, with a distance of 0.25 grid unit between the inflow plane and the plate leading edge at x = 0. The C-mesh around the NLF(1)-0416 airfoil has an extent of approximately 1000 chord lengths. The prolate spheroid is modeled as a half body in a domain of about 160 body lengths. The CRM-NLF mesh is comprised of 10 overset grids for the fuselage nose cap, fuselage front part, fuselage aft part, wing bottom, wing top, wing-body collar bottom, wing-body collar top, wing tip side cap, wing tip forward cap, wing tip aft cap. The farfield grids that are automatically generated within the solver. The reference chord length is 14.342 grid units (inches), and the length of the domain is 12,000 *in*, or approximately 100 fuselage lengths.

Table 2 Grid size information

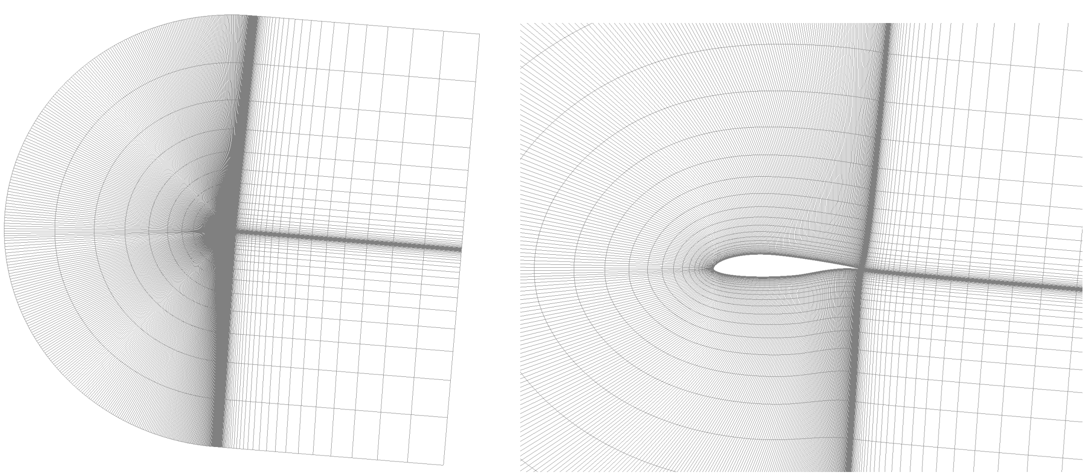
Case	Tiny	Coarse	Medium	Fine	Extra fine	Ultra fine
Flate plate	3,375	13,083	51,507	204,387	814,275	
NLF(1)-0416	51,891	115,851	205,155	459,795	815,811	1,831,971
Prolate Spheroid		545,025	1,815,937	4,276,737	14,340,865	
CRM-NLF (pre-blanking)		18,796,893	38,972,996	97,286,478		



(a) Overview

(b) Zoom near the plate leading edge

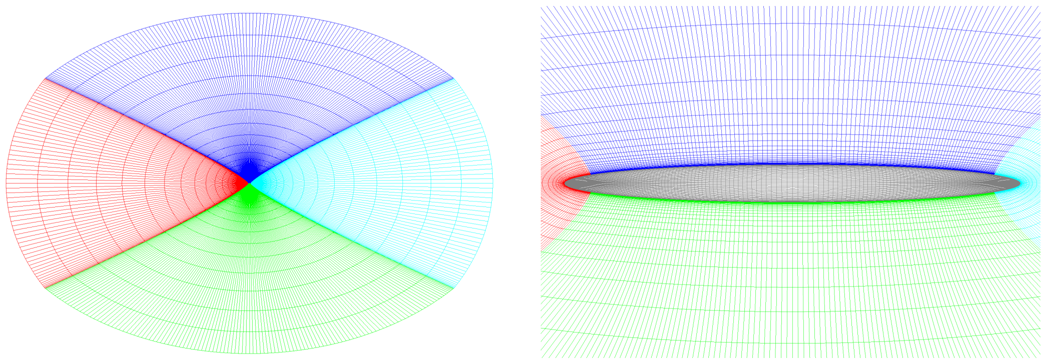
Fig. 1 Coarse grid for the flat plate



(a) Overview

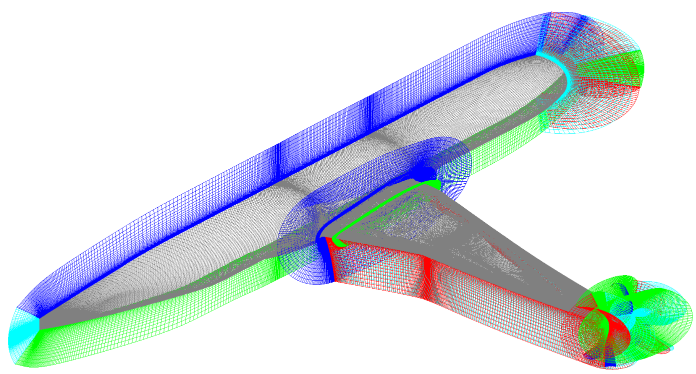
(b) Grid around the airfoil

Fig. 2 Coarse grid for the NLF(1)-0416 airfoil

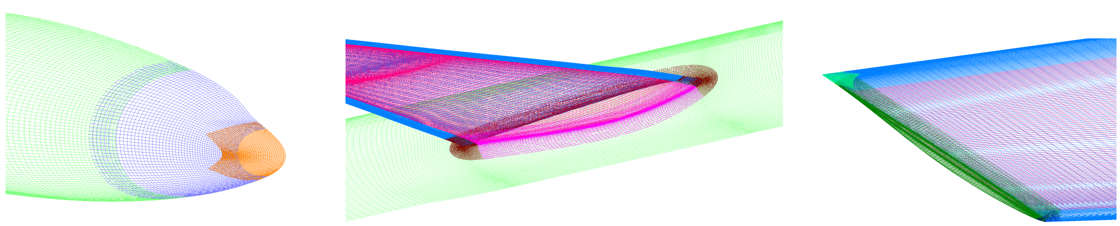


(a) Overview (b) Mesh on the half spheroid and in the symmetry plane

Fig. 3 Coarse grid for the prolate spheroid



(a) Body-fitted grid system. Volume mesh boundaries are shown with different colors.



(b) Surface grids at the nose of the fuselage (c) Bottom view of the wing mesh showing the overlap regions between lower and upper grids (d) Surface mesh at the wing tip showing the end cap grids

Fig. 4 Coarse grid for the CRM-NLF model

IV. Numerical Methods

A. OVERFLOW Solver

The version 2.3d of the node-centered overset curvilinear RANS computational fluid dynamics code OVERFLOW was used in this work. A 3rd-order MUSCL scheme with Roe upwind fluxes was used and the pseudo-time was advanced towards steady state using the improved implicit symmetric successive overrelaxation (SSOR) algorithm described in

Ref. [17]. The 2003 version of the SST turbulence model [3] was used in combination with the 2009 version of the Langtry-Menter model [2]. We label this combination SST-2003-LM-2009 model, following the nomenclature from the Turbulence Modeling Resources (TMR) homepage [18]. The version of the SST model with sustaining source terms (SST-2003-sus) was used for all cases but the flat plate cases, to avoid excessive decay of freestream turbulence intensity between domain inlet and wall [19]. The model extension for crossflow induced transition by Langtry [11] was used for the 3D inclined prolate spheroid. The transport equations for the turbulence kinetic energy k, specific dissipation rate ω, and transition γ and $\mathrm{Re}_{\theta t}$ variables are solved sequentially in a weakly coupled system. Viscous adiabatic boundary conditions are used at the walls with pressure extrapolation. A symmetry boundary condition is set upstream from the flat plate. The CFL number was limited to 10 with 5 turbulence model iterations per flow solver iteration. The farfield boundaries for the Navier-Stokes variables are treated as characteristic outflow condition based on Riemann invariants with freestream imposed on incoming characteristics.

B. Turbulence Index

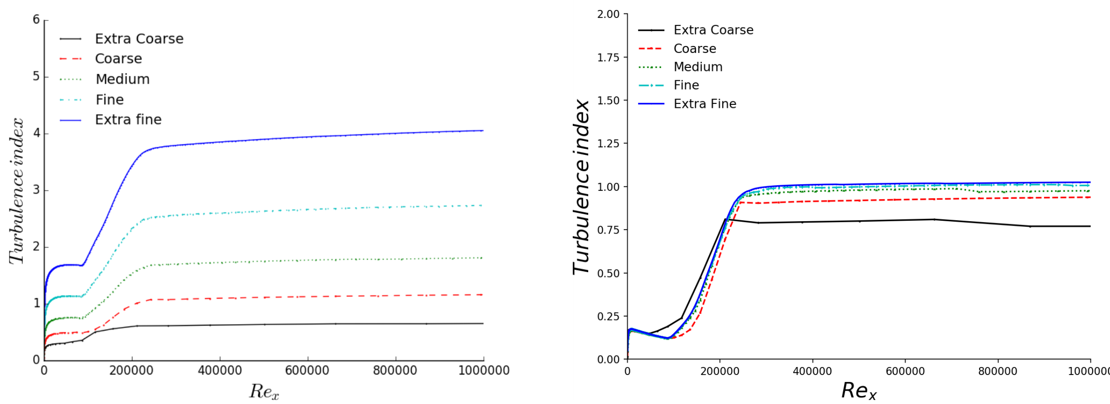
The turbulence index $\mathrm{i_t}$ is a convenient indicator of the location of transition. It is defined in such way that a value of 0 indicates laminar flow and a value of 1 indicates turbulent flow. It is common practice to use $\mathrm{i_t} = 0.5$ as a threshold to detect the location of a transition front. For the SST model, Strelets [20] proposed the following formulation that was implemented in the OVERFLOW solver:

$$i_t = 7.4 \frac{1}{u_\tau^{1.62}} \frac{d(\nu k^{0.31})}{dy} \tag{1}$$

In this equation, u_τ is the friction velocity and ν is the laminar kinematic viscosity. As reported by the author at the workshop, this coefficient proved unreliable when combined with the Langtry-Menter model: it is not well bounded within 0 and 1 and shows unphysical scaling with grid refinement, as illustrated in Fig. 5(a) for the flat plate case T3A. As a result, the location of transition fronts based on that index was inaccurate and inconsistent. Meanwhile, Carnes and Coder recalibrated the SST-based index to account for the modification of the turbulence kinetic energy near-wall scaling caused by coupling with the Langtry-Menter model [21]:

$$i_t = 6.1 \frac{1}{u_\tau^{2.346}} \frac{d(\nu k^{0.673})}{dy} \tag{2}$$

Fig. 5(b) shows the index profiles for this new version, which was used in the present work with the detection threshold $\mathrm{i_t} = 0.5$.



(a) Index per Eq. (1)

(b) Index per Eq. (2)

Fig. 5 Turbulence index along the flat plate (T3A case)

C. Freestream Turbulence Decay

In the SST model, the turbulence kinetic energy decay in the freestream can be expressed as [22]

$$\frac{k}{k_\infty} = \left(1 + \frac{\beta^*(C_{\epsilon 2} - 1)\omega_\infty x}{u_\infty}\right)^{-\frac{1}{C_{\epsilon 2}-1}} \tag{3}$$

and similarly for the specific turbulence dissipation rate

$$\frac{\omega}{\omega_\infty} = \left(1 + \frac{\beta^*(C_{\epsilon 2} - 1)\omega_\infty x}{u_\infty}\right)^{-1} \tag{4}$$

where $\nu_{\mathrm{t}\infty}$ is the freestream eddy viscosity, $\mathrm{C}_{\epsilon 2} = 1.92$, and β^* is a coefficient in the destruction term of the k transport equation. The farfield turbulence kinetic energy k_∞ is typically derived from turbulence intensity measurements with $\mathrm{k}_\infty = 1.5\mathrm{Tu}^2\mathrm{u}_\infty^2$, while the farfield specific turbulence dissipation rate is calculated based on the eddy viscosity

$$\omega_\infty = \frac{k_\infty}{\nu_{t\infty}} \tag{5}$$

or using the mixing length $\mathrm{l_m}$ per Prandtl's Reynolds stress viscosity hypothesis [23, 24]

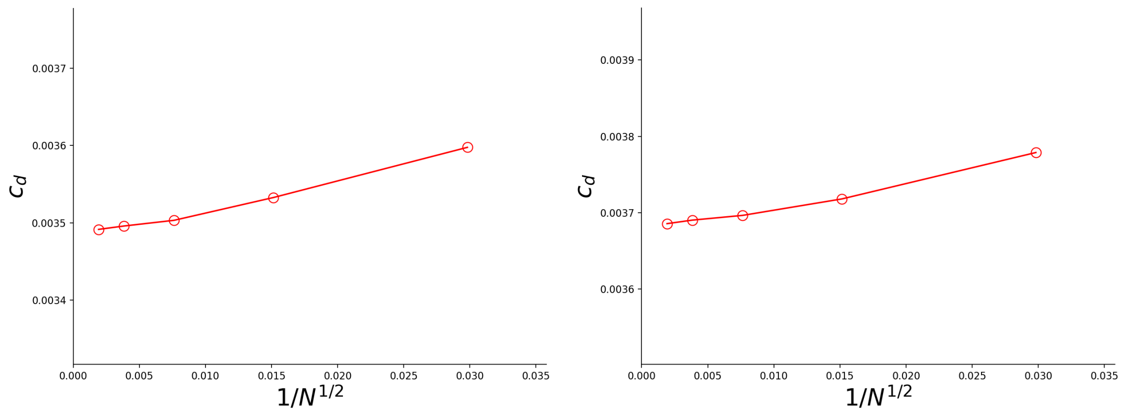
$$\nu_t = c\sqrt{k}\, l_m \tag{6}$$

where c = 0.55. In OVERFLOW, the turbulent viscosity ratio $\nu_{\mathrm{t}\infty}/\nu_\infty$ = MUTINF is provided as an inflow boundary condition. In absence of calibration data, $\nu_{\mathrm{t}\infty}$ and $\mathrm{l_m}$ are usually not known a-priori. Moreover, the SST model makes no claim of providing accurate modeling of the freestream decay. Meshing effects and bounds on k and ω to avoid unphysical alterations of the boundary layer solution may also interfere with a high-fidelity representation of freestream turbulence decay [19]. In a study of free decay using commercial software, Sarkar [25] determined that a value of $\beta^* \approx 0.046$ would better fit experimental and LES data than the standard value of $\beta^* = 0.09$. For these reasons, an accurate account of freestream turbulence kinetic energy decay cannot be guaranteed.

An additional difficulty when specifying the turbulent model farfield boundary conditions arises for large external aerodynamic domains, as for example when simulating in free air a model that was tested in a wind tunnel [26, 27]. In such a case, excessive turbulence kinetic energy decay between the inflow boundary and the wall may delay transition. The SST-sus version of the SST model with sustaining source terms was proposed in Ref. [19] to address this issue. In this approach, the farfield values of the turbulent variables are transported to the boundary layer edge without decay. The SST-sus method was therefore used for the NLF(1)-0416, prolate spheroid and CRM-NLF cases with large aerodynamic domain. The experimental Tu level was used to set k_∞ for the NLF(1)-0416 airfoil and inclined prolate spheroid, whereas a decay over a reference length was allowed for the CRM-NLF. The turbulent viscosity $\nu_{\mathrm{t}\infty}$ was set so that $\omega_\infty \mathrm{L}/\mathrm{u}_\infty = 5$. For the flat plate, baseline simulations used the values prescribed for the workshop, namely $\nu_{\mathrm{t}\infty}/\nu_\infty = 11.9$ for the T3A case, and $\nu_{\mathrm{t}\infty}/\nu_\infty = 99$ for the T3B case. The farfield boundary conditions for each case are listed in Table 3.

Table 3 Farfield boundary conditions for the SST model

Case	$\mathrm{k}_\infty/\mathrm{u}_\infty^2$	MUTINF
T3A	5.14×10^{-3}	11.9
T3B	7.81×10^{-3}	99.0
NLF(1)-0416	3.37×10^{-6}	2.7
Prolate Spheroid	3.37×10^{-6}	4.6
CRM-NLF	1.0×10^{-6}	3.0



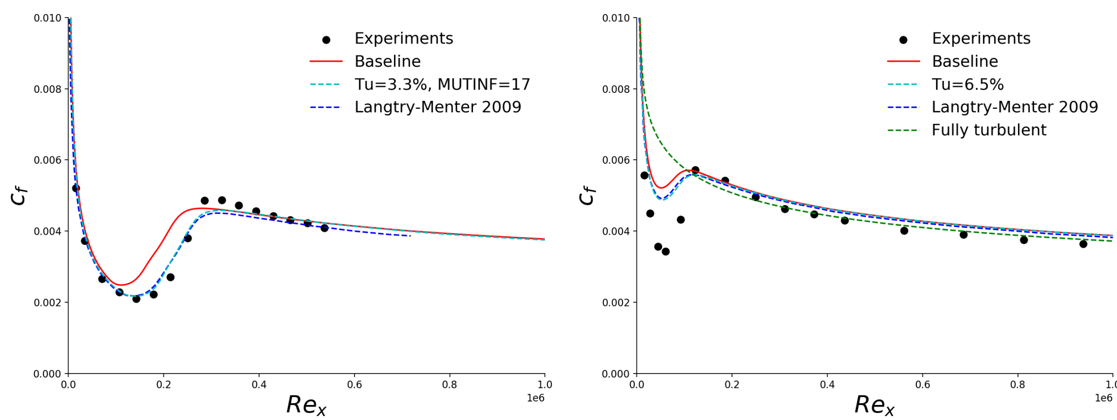
(a) T3A (b) T3B

Fig. 6 Drag coefficient versus mesh refinement for the T3A and T3B flat plate cases

V. Results

A. Flat plate

Fig. 6 shows that the drag coefficient for the T3A flat plate case asymptotically converges to $c_d = 3.495 \times 10^{-3}$, and the drag coefficient for the T3B case converges to $c_d = 3.685 \times 10^{-3}$. Measurements and simulations of the friction coefficient along the plate are shown in Fig. 7. Prior results from Ref. [2] are included, which were obtained using the same value of MUTINF as in Table 3, but slightly different inflow turbulence intensities, with Tu=3.3% for the T3A case, and T=6.5% for the T3B case. For the T3A case, the baseline simulation predicts transition upstream of the measured location. An improved alignment with the T3A measurements and Langtry's results is obtained by increasing the eddy to laminar viscosity ratio from 12 to 17. For the T3B case, a decrease of the baseline freestream turbulence intensity from 7.216% to 6.5% yields good alignment with Langtry's results, still over-predicting friction in the transition region.

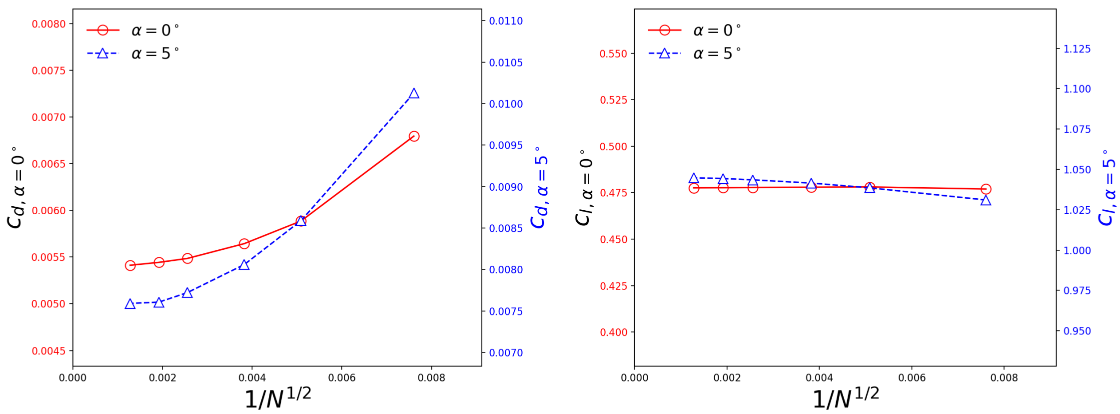


(a) T3A (b) T3B

Fig. 7 Friction coefficient along the flat plate, with measurements by Roach and Brierley [12] and prior simulations by Langtry [2]. Present simulations are shown for the baseline farfield boundary conditions in Table 3 (red) and for modified inflow boundary conditions (cyan). SST fully turbulent results are shown for the T3B case.

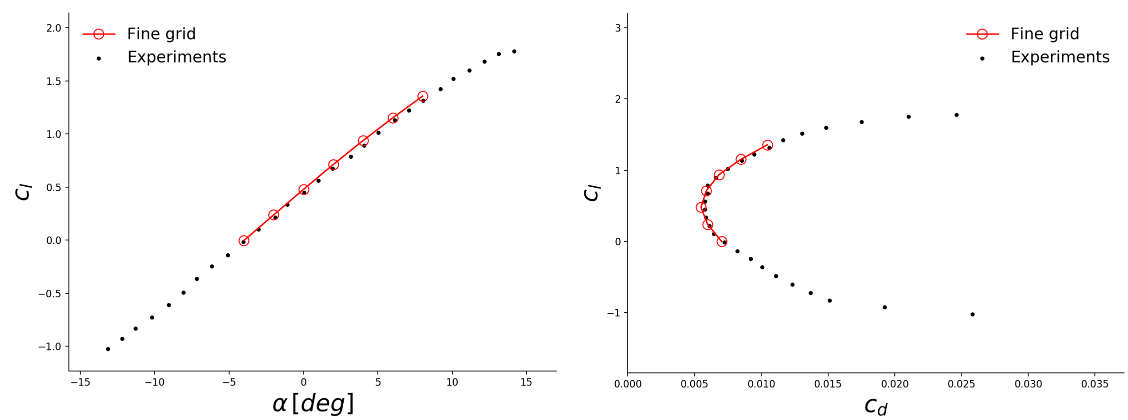
B. NLF(1)-0416 airfoil

Fig. 8 shows good asymptotic convergence of the drag and lift coefficients with mesh refinement for the NLF(1)-0416 airfoil. The lift coefficient is plotted in Fig. 9 for the fine grid. Good agreement with the experiments by Somers [13] is obtained. The pressure coefficient distributions at $\alpha = -2°$ and $\alpha = 8°$ are shown in Fig. 10 for the finest grid and are consistent with prior results [28]. The lift coefficient is plotted against the predicted transition location in Fig. 11, along with experimentally determined lower and upper bounds. Results are shown for the fine grid, they do not change significantly for the ultra-fine grid. On the upper side of the airfoil, the simulations overlap with the measurements improves at lower angles of attack, whereas the opposite is true for the lower side.



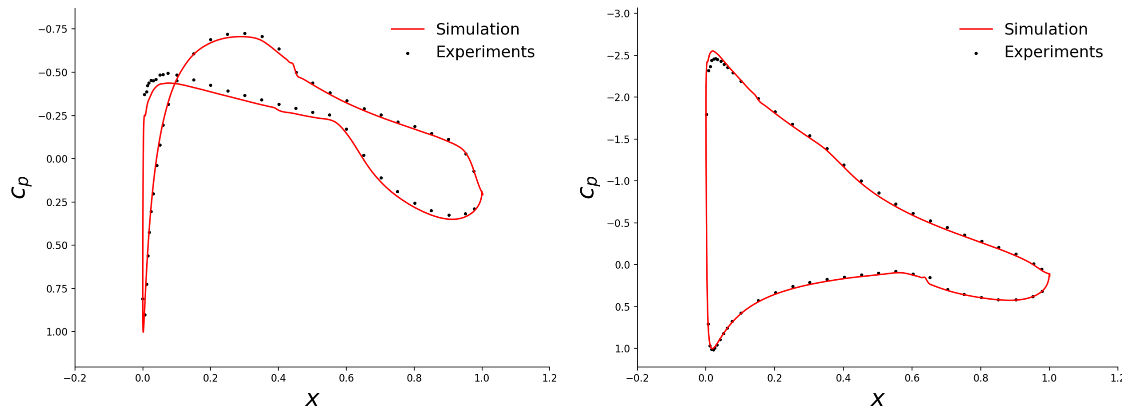
(a) Drag (b) Lift

Fig. 8 NLF(1)-0416 load coefficients versus grid refinement



(a) Angle of attack sweep (b) Drag Polar

Fig. 9 NLF(1)-0416 lift coefficient for the fine grid with experiments by Somers [13]



(a) $\alpha = -2°$

(b) $\alpha = 8°$

Fig. 10 NLF(1)-0416 pressure coefficient for the fine grid compared with experiments

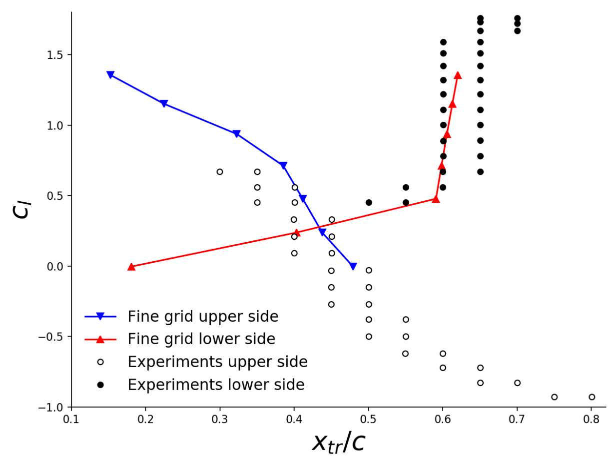
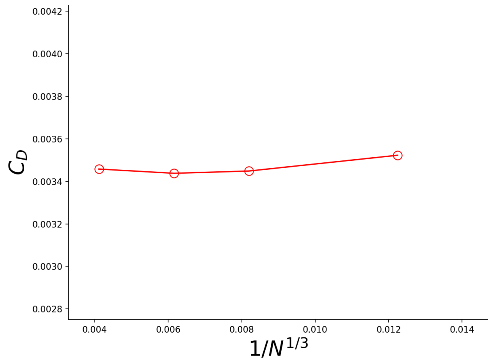


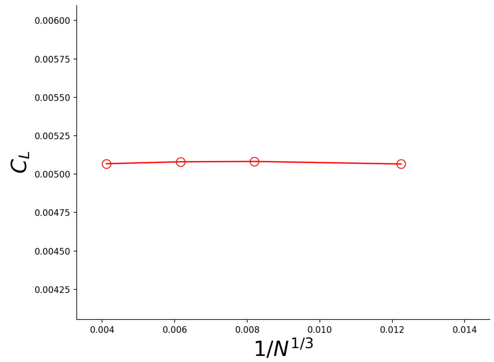
Fig. 11 Transition streamwise coordinate on the upper and lower sides of the NLF(1)-0416 airfoil per measurements in Ref. [13], and OVERFLOW simulations with the Langtry-Menter model on the fine grid. For each lift value, the measurements show the last probed position where the flow was laminar, and the first probed position where the flow was turbulent, providing bounds within which transition took place.

C. Inclined 6:1 prolate spheroid

The inclined prolate spheroid is a benchmark test case for the analysis of crossflow induced transition over a body akin to an airplane fuselage. The model extension for crossflow induced transition by Langtry [11] is used with an RMS roughness height of 3.3μm, as assumed in Ref. [2]. The baseline Tu of 0.15% is slightly higher than that of 0.1% considered by Langtry, and both values are within the range reported for the test. In Fig. 12 the load coefficients do not vary much with refinement but the drag coefficient shows a trend change between medium and extra fine grids, an additional refinement level would be needed to determine the asymptotic value. In Fig. 13, the location of transition is mapped using its angular coordinate ϕ around the streamwise axis, with $\phi = 0°$ at the bottom of the spheroid. The simulations with baseline settings predict transition downstream of the measurements on the lower part of the spheroid. Increasing the RMS roughness by a factor of up to 6, corresponding to an (unrealistic) RMS roughness height of 20μ m, pulls the transition front upstream on the upper side of the spheroid, and close to the locus predicted in Ref. [11] for $\alpha = 15°$ (Fig. 13(c)). However, this setting still does not capture the transition front on the lower side of the body. In Ref. [26], somewhat better results were obtained at low ϕ using a lower eddy viscosity ratio $\nu_{t\infty}/\nu_\infty = 1.2$ with sustaining terms. In the present case, increasing the eddy viscosity ratio to 100 did not improve the results significantly.

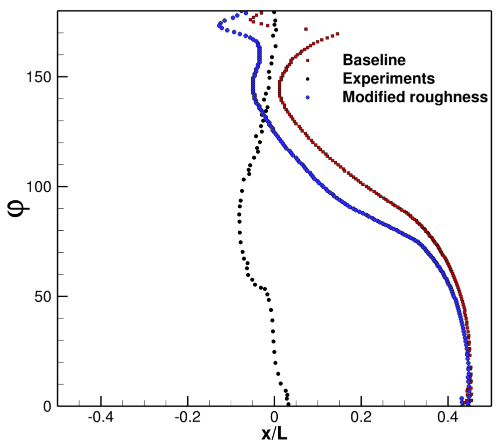


(a) Drag

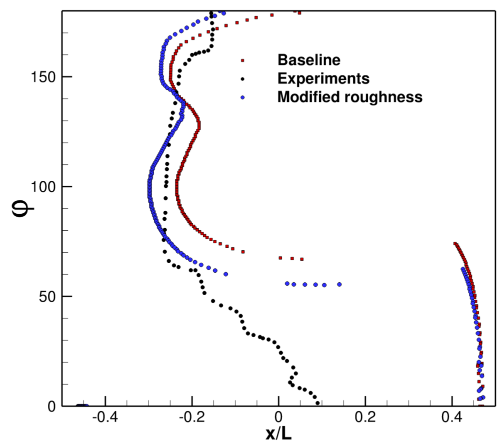


(b) Lift

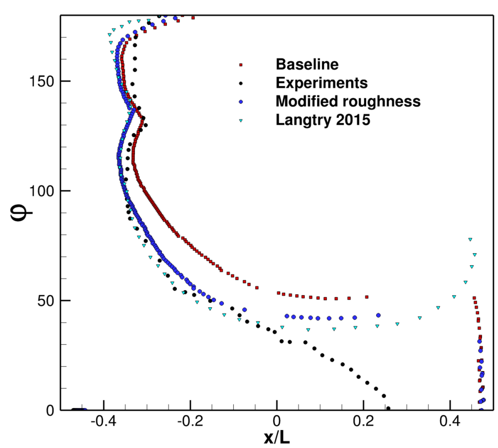
Fig. 12 Load coefficients versus mesh refinement for the 6:1 inclined prolate spheroid



(a) α = 5°



(b) α = 10°

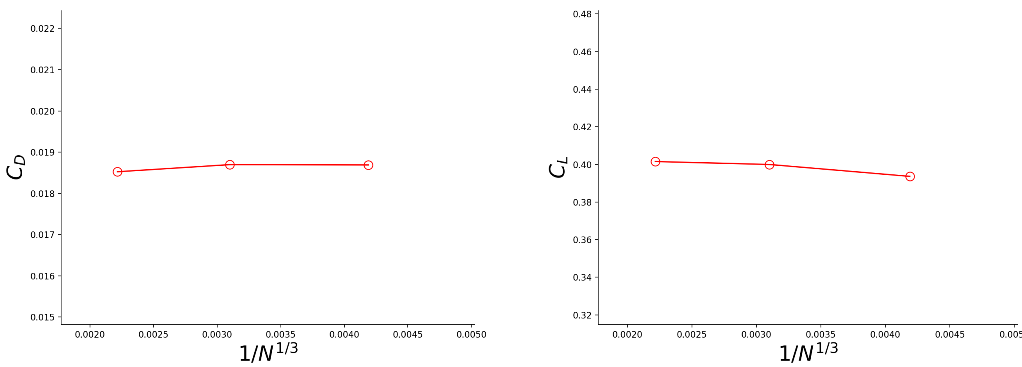


(c) α = 15°

Fig. 13 Locus of the transition location on the inclined prolate spheroid at different angles of attack (finest mesh).

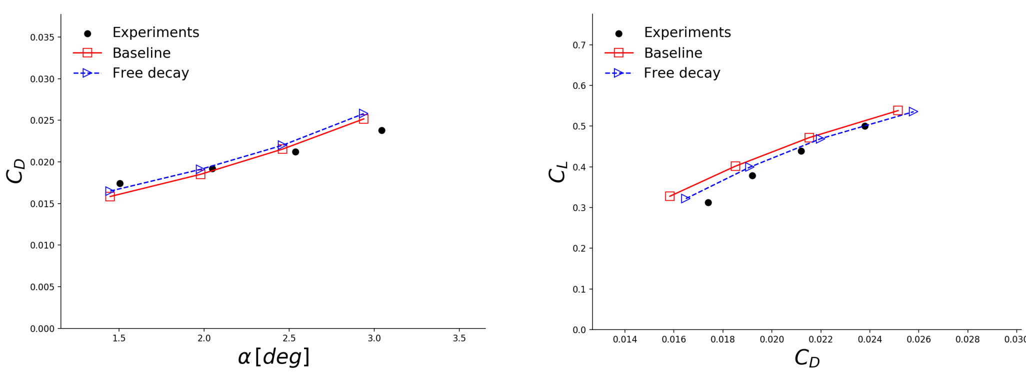
D. CRM-NLF

In Fig. 14 the drag coefficient of the CRM-NLF at $\alpha = 1.98°$ shows a slight downward trend between the medium and fine grids, whereas the lift coefficient slightly increases with refinement. An additional refinement level would be useful to confirm these trends. Load measurements [16] and computations for the fine grid are compared in Fig. 15. Results are shown for the baseline farfield boundary setup, and for modified boundary conditions using the experimental turbulence intensity Tu=0.24% with MUTINF=26 without sustaining source terms, so that the turbulence variables were allowed to decay between the farfield boundary and the wall. The simulated drag and lift coefficients are more sensitive to the angle of attack than in the experiments, and the lift coefficient is consistently higher. The drag for the free decay case is slightly larger than for the baseline. The transition lines simulated for the baseline case at the four angles of attacks are plotted over the Temperature Sensitive Paint (TSP) measurements in Fig. 16. Transition wedges emanating from the leading edge can be seen in the TSP. They likely result from surface imperfections or dust causing bypass transition. This hypothesis is supported by the decrease of the number of wedges observed after the wing surface was re-polished between wind tunnel runs Ref. [15]. The simulations, on the other hand, assume a perfectly smooth surface. Overall, the computed transition fronts are well aligned with the experimental ones at 2.46° and 2.94°, and along the outer third of the wing at the lower angles of attack, with transition predicted too far upstream on the inboard wing section. In the free turbulence decay case at $\alpha = 1.98°$ (Fig. 17(a)), better results are obtained along the inboard part of the wing, although transition is still predicted upstream of the measured front (Fig. 16(b)). Fig. 16(b) shows that when the sustaining terms are switched on while keeping the experimental Tu level at the inflow boundary, transition takes place close to the wing leading edge. Good agreement is found between measured and simulated pressure coefficients at $\alpha = 1.98°$, as shown for selected spanwise stations in Fig. 18.



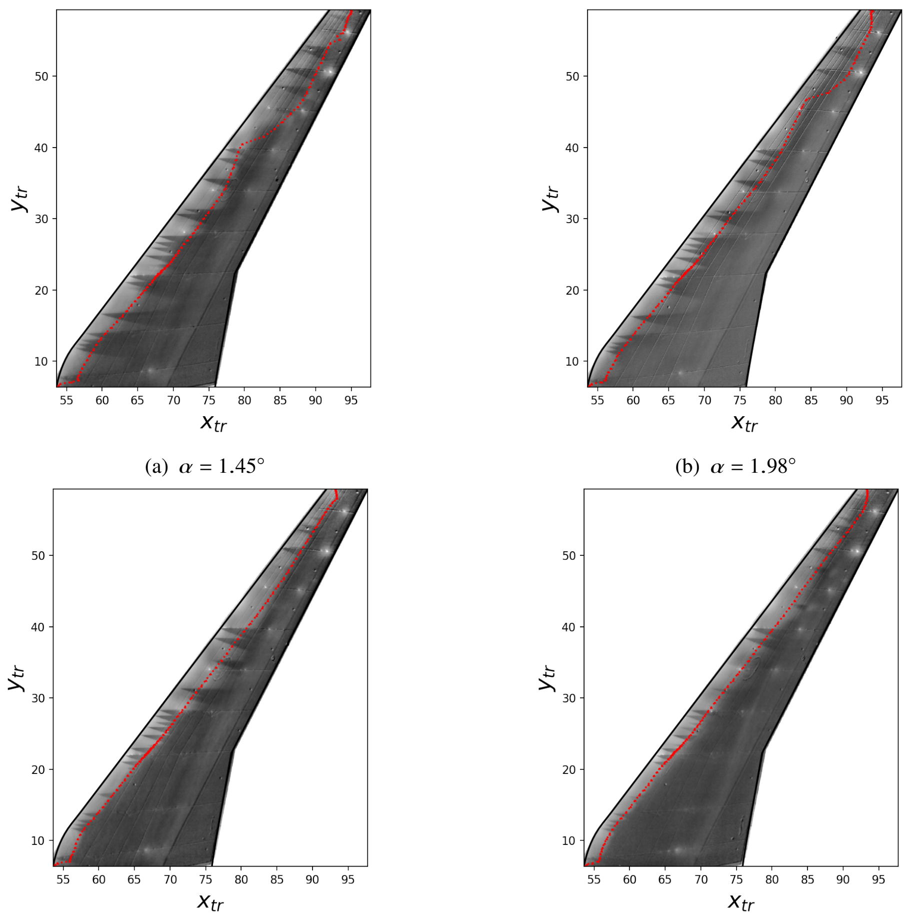
(a) Drag (b) Lift

Fig. 14 Load coefficients versus grid refinement for the CRM-NLF configuration at $\alpha = 1.98°$



(a) Drag (b) Drag polar

Fig. 15 Load coefficients for the CRM-NLF at different angles of attack



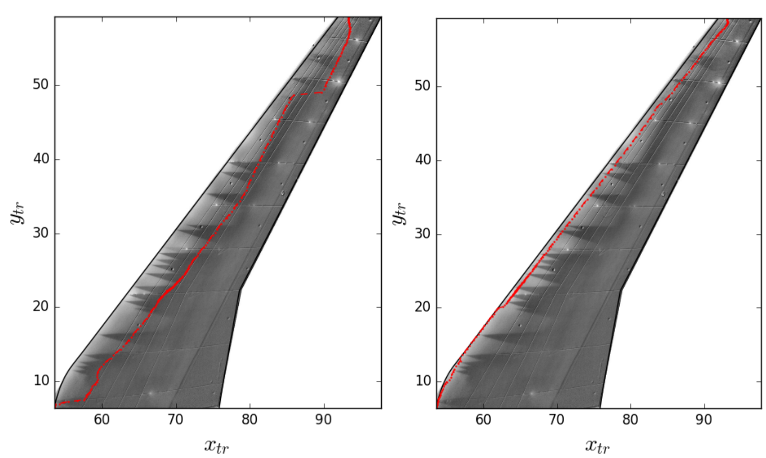
(a) α = 1.45°

(b) α = 1.98°

(c) α = 2.46°

(d) α = 2.94°

Fig. 16 Simulated transition line (red) superimposed over TSP maps of the CRM-NLF wing at different angles of attack



(a) Free turbulence decay

(b) Sustaining source terms

Fig. 17 Simulated transition line (red) for Tu=0.24% at the farfield boundary, plotted over TSP maps of the CRM-NLF (α = 1.98°)

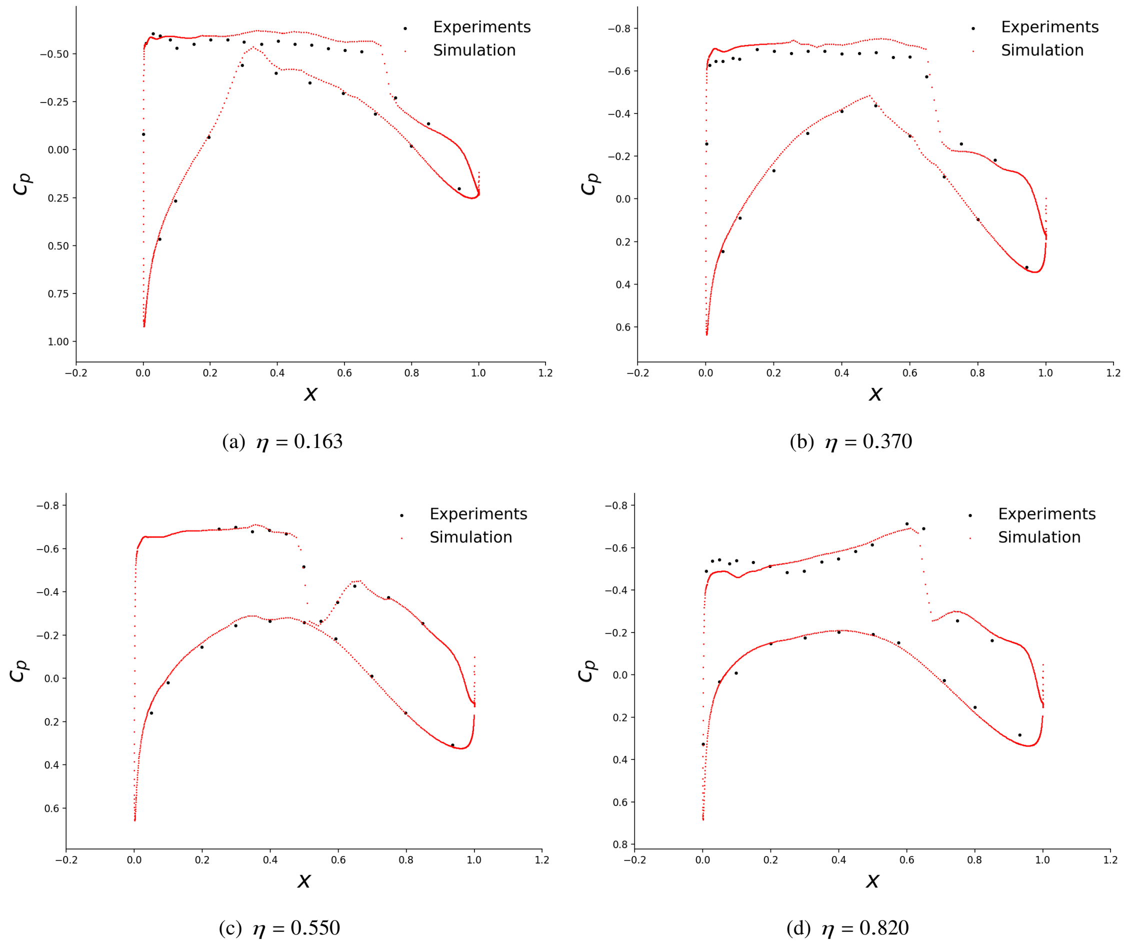


Fig. 18 Pressure coefficients for the CRM-NLF at $\alpha = 1.98°$ at different spanwise locations $\eta = y/S$, where S is the wing span

VI. Conclusions

The Shear Stress Transport (SST) turbulence model and the Langtry-Menter transition model were applied to the test cases of the 1st AIAA CFD Transition Modeling and Prediction Workshop using the OVERFLOW solver. For the flat plate case T3A, the baseline farfield boundary conditions led to transition upstream of the measured location. An adjustment of the turbulence intensity and an increase of the eddy viscosity led to improved alignment with experiments and earlier simulation results. For the T3B case, the simulations aligned well with prior results using the prescribed eddy to laminar viscosity ratio and the experimental turbulence intensity. The results for the NLF(1)-0416 airfoil were in good agreement with the experiments. Transition on the prolate spheroid was predicted downstream of the measured location on the lower part of the body. An increase of the RMS roughness height yielded a better agreement with earlier simulations at $\alpha = 15°$ but still led to transition too far downstream at low ϕ, in spite of using higher freestream turbulence levels than comparable benchmark. For the CRM-NLF configuration, the load coefficients showed higher sensitivity to the angle of attack than in the experiments. Good transition predictions were obtained at the higher angles of attack. When prescribing the experimental Tu at the farfield boundary, free turbulence decay improved the predicted transition location on the inboard side of the wing. Activation of the SST sustaining source terms led to transition near the leading edge.

Except for the NLF(1)-0416 airfoil, modifications of the initial boundary settings helped improve the results, but did not yield complete agreement with the measurements. The present findings illustrate the need for improved best practices to set the farfield boundary conditions of the SST model when it is coupled with the Langtry-Menter model. Experimental characterization of turbulence decay and surface roughness of highly swept geometries in future transition test campaigns would be helpful to isolate model shortcomings from model input uncertainties. The model has already shown capabilities to account for a large range of transition cases with appropriate adjustment of its input parameters. Addressing the concerns raised here and by other authors may advance these capabilities to reliably predict laminar-turbulent transition in unknown configurations.

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